Exp No:20 Experiment to understand K-means clustering algorithm for a given data set.

AIM:

To implement the k-means clustering algorithm on a given dataset.

Description:

Understand the K-means clustering algorithm for the dataset given.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
df=pd.read_csv('Mall_Customers.csv')
df.info()
df.head()
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
```

```
plt.show()

features_el=df.iloc[:,[2,3,4]].values

from sklearn.cluster import KMeans

wcss=[]

for i in range(1,10):

model=KMeans(n_clusters=i)

model.fit(features_el)

wcss.append(model.inertia_)

plt.plot(range(1,10),wcss)
```

OUTPUT:

1. Dataset Information

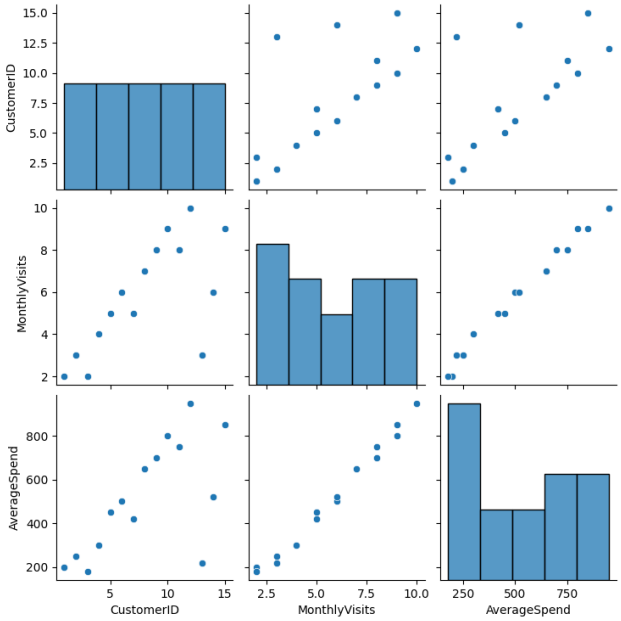
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   CustomerID    15 non-null    int64
 1   MonthlyVisits 15 non-null    int64
 2   AverageSpend  15 non-null    int64
dtypes: int64(3)
memory usage: 492.0 bytes
```

2. Original Dataset – First Five Records

	CustomerID	MonthlyVisits	Average Spend
0	1	2	200
1	2	3	250
2	3	2	180
3	4	4	300

4	5	5	450

3. Pair Plot



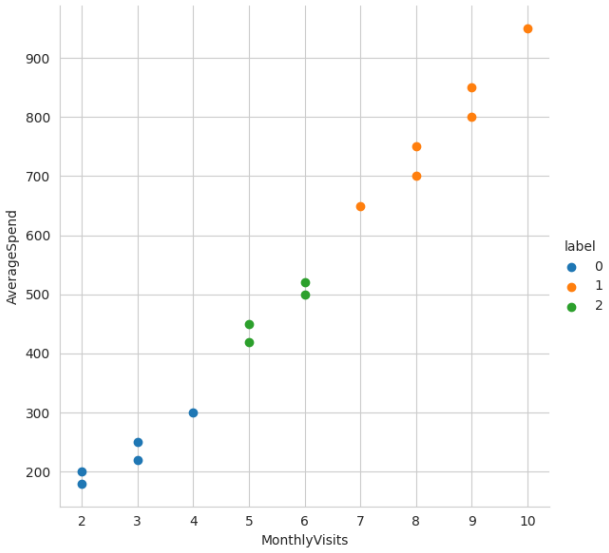
4. K-Means Model

KMeans(n_clusters=3, n_init=10, random_state=42)

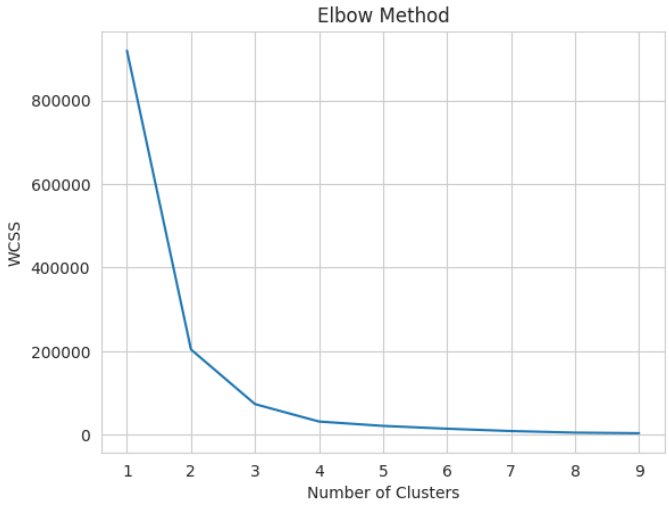
5. Clustered Data – First Five Records

	MonthlyVisits	AverageSpend	label
0	2	200	0
1	3	250	0
2	2	180	0
3	4	300	0
4	5	450	2

6. Cluster Visualization



7. Elbow Method – WCSS



8. Final K-Means Model

KMeans(n_clusters=3, n_init=10, random_state=42)

9. Final Clustered Dataset

	MonthlyVisits	AverageSpend	label
0	2	200	0
1	3	250	0
2	2	180	0
3	4	300	0
4	5	450	2
5	6	500	2
6	5	420	2

7	7	650	1
8	8	700	1
9	9	800	1
10	8	750	1
11	10	950	1
12	3	220	0
13	6	520	2
14	9	850	1

10. Final Cluster Output

Customer groups are represented by the labels 0, 1 and 2.
The final dataset shows the cluster assigned to each customer based on MonthlyVisits and AverageSpend.

RESULT:

The café customers were successfully grouped into 3 clusters using k-means clustering.